Cheat Sheet: Which Differentiation Rule Do I Use?

How to use this sheet

Same idea as the integration decision tree — work through these questions in order. Many problems need more than one rule stacked together, so don't stop at the first "yes" if there's still more structure left to handle inside that piece.

Step 1: Is it a sum or difference of separate terms?

$$\text{Example: } y = x^3 + \sin x - e^x$$

→ Differentiate each term completely separately, then add/subtract the results. Handle each individual term using the rest of this guide.

Step 2: Look at one term at a time. Is it two different functions multiplied together?

(Not a constant times a function — an actual product of two functions that both involve x.)

$$\text{Examples: } y = x^2 \sin x \qquad y = xe^x \qquad y = \ln x \cdot x^3$$

→ Product Rule: $\dfrac{d}{dx}[fg] = f'g + fg'$

Before reaching for this: check if the product would be easier to just expand out first (like $x(x^2-3)$) — if so, expand and skip the product rule entirely.

Step 3: Is it one function divided by another?

$$\text{Example: } y = \frac{\sin x}{x} \qquad y = \frac{x^2}{x+1}$$

→ Quotient Rule: $\dfrac{d}{dx}\left[\dfrac{f}{g}\right] = \dfrac{f'g - fg'}{g^2}$

Before reaching for this: check if the fraction simplifies by dividing term-by-term (like $\dfrac{x^2+4x}{x} = x+4$) — if so, simplify first and use the power rule instead. It's almost always faster.

Step 4: Is there a "function wrapped inside another function"?

(The output of one function is being fed directly into another — a composition.)

$$\text{Examples: } y = \sin(x^2) \qquad y = e^{3x} \qquad y = (x^2+1)^5 \qquad y = \ln(\cos x) \qquad y = \sqrt{x^3+2}$$

$\rightarrow$ Chain Rule: identify the "outer" function and the "inner" function. Differentiate the outer function (leaving the inside alone), then multiply by the derivative of the inside.

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

Step 5: Does the problem combine several of the above?

This is extremely common — don't be surprised when a single term needs two or three rules layered together.

Product + Chain, example: $y = x^2 \sin(3x)$

$\rightarrow$ Use the Product Rule first (treating x^2 and $\sin(3x)$ as the two factors). While differentiating the $\sin(3x)$ piece, you'll need the Chain Rule (since the inside is $3x$, not plain x).

Quotient + Chain, example: $y = \dfrac{e^{2x}}{x+1}$

$\rightarrow$ Use the Quotient Rule first. While finding the derivative of e^{2x} (the numerator), you'll need the Chain Rule.

Step 6: Is y defined implicitly — mixed up with x in an equation, not solved for y by itself?

Example: $x^2y + xy^2 = 8$

$\rightarrow$ Implicit Differentiation. Differentiate both sides with respect to x, and remember: every single time you differentiate a term containing y, you must tack on a y' (chain rule, since y is secretly a function of x). Then collect all the y' terms on one side and solve for y'.

The special-function derivative reference

Function	Derivative
x^n	nx^{n-1}
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
e^x	e^x
$\ln x$	$\dfrac{1}{x}$
$\arcsin x$	$\dfrac{1}{\sqrt{1-x^2}}$
$\arctan x$	$\dfrac{1}{1+x^2}$

For any of these with something other than plain x inside (like $\sin(3x)$ or $\ln(x^2+1)$), multiply by the chain rule factor u'.